COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST

AIR FLOW TESTER

MODEL AF-R220

ERROR AND TROUBLESHOOTING MANUAL

1 Error

1.1 When an Error Occurred

The error code is displayed when an error occurs.

The error description, probable causes and the treatments are displayed by tapping the displayed code.

1.2 Error List

Menu to view descriptions, probable causes, and treatments and all the errors.

- **Go to**: Troubleshooting > Error List
- The error code is displayed when an error occurs.
  The errors are divided every 10 codes.
- Tapping ↑ ↓ goes forward or back within 10 errors.

2 Error Messages and Treatments

2.1 Error 1: PS Offset Error

Timing: During power-on check procedure. At the end of Charge Delay (DL1)

Criteria: Go to: Settings Menu > Advanced Settings > Test Press Pressure sensor (PS) offset exceeds $\pm 2\%$ of its range (TP LL).

Probable Cause	Treatment
Test pressure sensor (PS) offset is out of tolerance when the power is turned on and/or at the end of Charge Delay (DL1).	Adjust the PS offset. Go to: Main Menu > Inspection > Sensor > PS Contact Cosmo for repair if the PS offset exceeds $\pm 2\%$ of its range.

2.2 Error 2 PS Out of Range

Timing: At the end of Pressurization (CHG), Precharge (PCHG), L-DET and Detection (S-DET)

Criteria: Test pressure exceeds the sensor range

Probable Cause	Treatment
Test pressure sensor (PS) was pressurized exceeding the sensor full-scale.	Adjust the test pressure. Pay extra attention for low pressure models.
Test pressure sensor (PS) offset is out of tolerance.	Adjust the PS offset. Go to: Main > Inspection > Sensor > PS

	Contact Cosmo for repair if the PS offset exceeds $\pm 2\%$ of its range.
Cable disconnection or malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

2.3 ERROR 3 Improper Test Pressure

Timing:	**Test pressure too low:**	At the end of Pressurization (CHG), L-DET and Detection (S-DET). (Settings Menu > Advanced Settings > Test Press > It depends on the setting of Pressure Monitoring.)
	Test pressure too high:	Always monitored during CHG, L-DET and S-DET. (Settings Menu > Advanced Settings > Test Press > It depends on the setting of Pressure Monitoring.)
	Precharge pressure too low:	At the end of Precharge.
	Precharge Pressure too high:	Always monitored.

Criteria: Test pressure exceeds upper or lower limit in CHG or PCHG stage

Probable Cause	Treatment
Upper and lower limits for Test pressure or Precharge are too close or inappropriate.	Check the upper and lower limits for Test pressure or Precharge. For CHG Stage limits: Go to: Settings Menu > Advanced Settings > Test Press > Upper/Lower Press Limit For PCHG stage limits: Go to: Settings Menu > Advanced Settings > CHG Options > Upper/Lower Precharge Limit
Pressurization time is insufficient. (When pressure is lowered.)	Extend CHG timer. Go to: Settings Menu > Advanced Settings > Timer > Pressurization (CHG)
Precharge time is insufficient. (When Precharge pressure is lowered.)	Extend PCHG timer. Go to: Settings Menu > Advanced Settings > CHG Options > Precharge Timer (PCHG)
Fluctuation or a drop in the source pressure	Check the source pressure or the regulator setting. Avoid using air tools branching off the pressure source of the AF-R220 to supply stable air. Setting up a dedicated pressure source for the AF-R220 is recommended.
Leaks from gaskets, part and fittings	Check the gaskets and fittings for possible leaks.
Malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

NOTE

When the error occurred in L-DET or S-DET, also the stages shaded with diagonal lines are applicable.

2.4 ERROR 10 Flow Sensor Offset Error

Timing: During power-on check procedure. At the end of Charge Delay (DL1)

Criteria: For Laminar Flow Models, the Differential Pressure Sensor (DPS) offset exceeds 10% of its range (± 100 Pa).

In Mass Flow Models, the output of Mass Flow Sensor exceeds $\pm 10\%$ of its F.S. range.

Probable Cause	Treatment
Contamination by foreign substances such as water or oil at the time of power-on.	Adjust the Flow Sensor offset. Go to: Maint Menu > Inspection > Flow Calib > Flow Calib U Range Contact Cosmo for repair if the offset exceeds $\pm 10\%$ of F.S. flow range.

2.5 ERROR 14 Air Operated Valve Error 4

Timing: At the end of Air-Blow (BLW)

Criteria: The flow during Air-Blow did not reach the Blow Check Limit

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the AF-R220 to supply stable air. Setting up a dedicated pressure source for the AF-R220 is recommended.
Air-Blow (BLW) timer is too short, or Blow Check Limit is too high.	Extend the Air-Blow (BLW) timer or lower the Blow Check Limit. Air-Blow (BLW) Timer: Go to: Settings Menu > Advanced Settings > Timer > Air-Blow (BLW) Blow Check Limit: Go to: Settings Menu > Advanced Settings > Self Check > Blow Check Limit
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.6 ERROR 15 Air Operated Valve Error 5

Timing: At the end of Detection (S-DET) (Only for High press)

Criteria: Pressure switch monitoring the pilot pressure for the fill valve is not activated.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the AF-R220 to supply stable air. Setting up a dedicated pressure source for the AF-R220 is recommended.
Malfunction of the pressure switch monitoring the pilot pressure.	Contact Cosmo for repair. As a provisional measure, the pressure switch monitoring can be disabled. Go to: Settings Menu > Common Settings > Special > PSW Monitoring > Disable

2.7 ERROR 22 Stop Valves Closed

Timing: At the end of PCHK (When closed while the tester is on, at the end of each stage.)

Criteria: The stop valve monitoring switch is ON/OFF

Probable Cause	Treatment
The stop valve of WORK port is closed, or the stop valve of CAL port is open. For the option CX, the stop valves of WORK port and CAL port are closed.	Check the stop valves and close the cover.
If the Error occurs even though the cover is closed, the stop valve monitoring switch may be malfunctioned or wiring disconnection may be broken.	Contact Cosmo for repair. The stop valve monitoring can be disabled as a provisional measure. Go to: Settings Menu > Common Settings > Special > Stop Valve Monitoring > Disable

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±2 t When the error occurred in L-DET or S-DET, also the stages shaded with diagonal lines are applicable.
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2.8 ERROR 23 Fixed Value Comp Error

Timing: At the end of COMP

Criteria: In Fixed Comp Sampling, the flows in S-DET and COMP exceed the Upper or Lower Comp Limit. The S-DET timer and Comp timer are 0 (s).

Probable Cause	Treatment
Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) timer and/or Detection (L-DET) timer. Go to: Settings Menu > Advanced Settings > Timer > Pressurization (CHG) / Detection (L-DET)
Upper Comp Limit and/or Lower Comp Limit are inappropriate.	Set larger limits. Go to: Comp Menu > Fixed Value Comp > Basic > Upper Comp Limit / Lower Comp Limit
Detection (S-DET) timer and/or Comp Timer are 0.0 s.	Set Detection (S-DET) timer and/or Comp Timer. Go to: Settings Menu > Advanced Settings > Timer > Detection (S-DET) Go to: Comp Menu > Fixed Value Comp > Basic > Comp Timer

NOTE

When the error occurred in L-DET or S-DET, also the stages shaded with diagonal lines are applicable.

2.9 ERROR 26 Temperature Sensor Error

Timing: At the end of Detection (S-DET)
Criteria: The temperature is out of the range of 0 to 50 °C.

Probable Cause	Treatment
Temperature Sensor Malfunction	The Temperature Sensor needs to be replaced. Contact Cosmo for repair.

NOTE

When the error occurred in L-DET or S-DET, also the stages shaded with diagonal lines are applicable.

2.10 ERROR 27 Atmospheric Pressure Sensor Error

Timing: At the end of Detection (S-DET)
Criteria: The atmospheric pressure is out of the range of 800 to 1200 hPa.

Probable Cause	Treatment
Atmospheric Pressure Sensor Malfunction	The Atmospheric Pressure Sensor needs to be replaced. Contact Cosmo for repair.

2.11 ERROR 28 Flow Optimizer Error

Timing: At the end of Detection (S-DET)
Criteria: The test pressure exceeded $\pm 20\%$ of the target test pressure. The test pressure exceeded the upper press limit or lower press limit set for the sampling in Flow Optimization.

Probable Cause	Treatment

The Test pressure exceeded or dropped below the sampled value at the end of S-DET.	Treatment 1: Check the regulator pressure. Treatment 2: Perform Flow Optimizer Sampling again.

2.12 ERROR 51 to ERROR 61 System Errors

Usually system errors (ERROR 51 to ERROR 61) are caused by malfunction of electrical components.

NOTE ERROR 51 (Lo Battery SRAM Error) may occur because of the battery life.

Error Code	Error Message	Description
ERROR 51	Low Battery Error	Probable Cause 1: The battery has discharged completely. Treatment 1: If this error occurs, none of the measurements is enabled. Replace the battery instantly following the procedures in the manual (Maintenance). Probable Cause 2: If the same error occurs after replacing the battery, some electrical parts may be malfunctioned. Treatment 2: Contact Cosmo for repair after executing a System Backup.
ERROR 52	AD Communication Error	Some electrical parts inside the AF-R220 may be malfunctioned. See below for the treatments.
ERROR 53	I/O Communication Error	Some electrical parts inside the AF-R220 may be malfunctioned. See below for the treatments.
ERROR 60	SD Card Error	Some electrical parts inside the AF-R220 may be malfunctioned. See below for the treatments.
ERROR 61	SRAM Error	Some electrical parts inside the AF-R220 may be malfunctioned. See below for the treatments.

Reboot AF-R220 or tap Stop on a measurement screen to cancel the error message. Contact Cosmo for repair after executing System Backup.

Cancelling the Error Message

1) Switch the operation mode to Manual (M)

2) Tap Stop on a measurement screen to cancel the error message. Rebooting AF-R220 also cancels the error. Or go to: Main. > Reboot

3 Frequent Fails

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the AF-R220 is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

2 Check the fixture condition

Probable Cause	Treatment
Leaks from pipe fittings	Look for leaks in the fittings by performing a bubble test applying soap solution by pressurizing the part (CHG Hold). Redo the piping if needed.
Deformation of pipe	Replace the pipe with the harder one that does not deform with the air pressure.

Proceed to the next item to check if the problem persists without identifiable causes.

3 Check the sealing condition

Probable Cause	Treatment
Sealing material is missing.	Place the sealing material.
Sealing surface is contaminated.	Clean the sealing surface.
Sealing material is damaged or worn-out.	Replace it with a new one.
Sealing deforms when the fixture clamps.	Check the following: Whether there is enough clearance between the sealing material and the groove. Wear of the stopper. Whether the size and hardness of the sealing material are appropriate. If the thrust force of the cylinder has lowered.

Proceed to the next item to check if the problem persists without identifiable causes.

4 Check whether there were environmental changes

Probable Cause	Treatment
Tested part is exposed to the direct wind from air conditioner or fan.	Move the source of the wind to where the wind does not hit the tested parts directly.
Some air tools are branched off the pressure source for the AF-R220 causing fluctuation in the pressure source.	Avoid using air tools branching off the pressure source of the AF-R220 to supply stable air. Setting up a dedicated pressure source for the AF-R220 is recommended.
Air compressor capacity is insufficient.	Use the air compressor whose capacity is large enough.
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.

Proceed to the next item to check if the problem persists without identifiable causes.

5 Check the condition of the tested parts

Probable Cause	Treatment
Part temperature is higher or lower than the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line.
The tested parts are wet.	If the part is wet, add or improve the drying process. Improve the drying process or add one if there isn't any.
The tested parts get deformed by pressurization.	Add a stopper to prevent the deformation.
Leak due to the gas porosity or internal leak.	Look for leaks in the fittings by performing a bubble test applying soap solution. If no leak is confirmed, there may be internal leaks. If there is a leak, re-evaluate the production process.